Lecture Note: The Divergence of Vector Fields

VACV Sem IV-Jan May 2026 (Lec 9)

1 Formal Definition

Let $U \subseteq \mathbb{R}^3$ be an open set. Let $\mathbf{F} : U \to \mathbb{R}^3$ be a differentiable vector field defined by component functions:

$$\mathbf{F}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k} \tag{1}$$

where $P, Q, R \in C^1(U)$ (continuously differentiable scalar functions).

The **divergence** of $\mathbf{F}$, denoted div $\mathbf{F}$ or $\nabla \cdot \mathbf{F}$, is the scalar function defined by [1]:

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \tag{2}$$

2 Classification of Divergence Behaviors

We categorize the behavior of the field based on the sign of the scalar field $\nabla \cdot \mathbf{F}$.

2.1 Case 1: Positive Divergence ($\nabla \cdot \mathbf{F} > 0$)

If $\nabla \cdot \mathbf{F} > 0$ at a point p, p is a **source** of scalar flux.

- **Example:** Let $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.
- **Calculation:**
$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$$
- Since $3 > 0$ everywhere, the field represents a uniform expansion of volume elements.

2.2 Case 2: Negative Divergence ($\nabla \cdot \mathbf{F} < 0$)

If $\nabla \cdot \mathbf{F} < 0$ at a point p, p is a **sink** of scalar flux.

- **Example:** Let $\mathbf{F}(x, y, z) = -x\mathbf{i} - y\mathbf{j} - z\mathbf{k}$.
- **Calculation:**
$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(-x) + \frac{\partial}{\partial y}(-y) + \frac{\partial}{\partial z}(-z) = -3$$
- Since $-3 < 0$, the vector field compresses volume elements at every point in the domain.

2.3 Case 3: Zero Divergence ($\nabla \cdot \mathbf{F} = 0$)

If $\nabla \cdot \mathbf{F} = 0$ everywhere in the domain, $\mathbf{F}$ is termed a **solenoidal vector field**.

- **Example:** Let $\mathbf{F}(x, y, z) = -y\mathbf{i} + x\mathbf{j} + 0\mathbf{k}$.
- **Calculation:**
$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(-y) + \frac{\partial}{\partial y}(x) + \frac{\partial}{\partial z}(0) = 0 + 0 + 0 = 0$$
- There is no net expansion or compression of volume elements; the flow is volume-preserving.

[1] the trace of the Jacobian matrix of $\mathbf{F}$

3 Physical Intuition: The Fluid Analogy

Imagine the vector field $\mathbf{F}$ represents the velocity flow of a fluid (liquid or gas). The divergence at a specific point (x, y, z) tells us whether fluid is being created (sourced) or destroyed (drained) at that point.

- **Flux Interpretation:** Divergence is the net flux per unit volume.
- If we draw a tiny box around a point, divergence measures:

$$\text{(Rate fluid flows out)} - \text{(Rate fluid flows in)}$$

4 The Three Cases of Divergence

Case 1: Positive Divergence ($\nabla \cdot \mathbf{F} > 0$)

Intuition: The "Source".

- Imagine a faucet turned on inside a tub of water, or a compressed gas canister exploding.
- There is a net flow *outward* from the point. The fluid is expanding.

Example: $\mathbf{F} = x\hat{i} + y\hat{j} + z\hat{k}$ (The position vector).

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$$

Since $3 > 0$, every point acts as a source, pushing material outward.

Case 2: Negative Divergence ($\nabla \cdot \mathbf{F} < 0$)

Intuition: The "Sink".

- Imagine a drain in a bathtub or a vacuum cleaner sucking in air.
- There is a net flow *inward* toward the point. The fluid is compressing.

Example: $\mathbf{F} = -x\hat{i} - y\hat{j} - z\hat{k}$.

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(-x) + \frac{\partial}{\partial y}(-y) + \frac{\partial}{\partial z}(-z) = -1 - 1 - 1 = -3$$

Since $-3 < 0$, every point acts as a sink, gathering material inward.

Case 3: Zero Divergence ($\nabla \cdot \mathbf{F} = 0$)

Intuition: "Incompressible" or "Solenoidal".

- The amount of fluid entering a tiny region exactly equals the amount leaving it.
- Examples include magnetic fields ($\nabla \cdot \mathbf{B} = 0$) or water flowing through a pipe without changing density.

Example: $\mathbf{F} = -y\hat{i} + x\hat{j}$ (Rotational/Vortex field).

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(-y) + \frac{\partial}{\partial y}(x) = 0 + 0 = 0$$

The fluid circulates, but it does not expand or compress.

5 The Divergence of the Gravitational Force Field

We examine the gravitational force field $\mathbf{F}$ defined on the domain $\Omega = \mathbb{R}^3 \setminus \{(0,0,0)\}$.

$$\mathbf{F}(\mathbf{r}) = -\frac{GMm}{r^2}\hat{r} \tag{3}$$

where G, M, m are positive constants, $r = \|\mathbf{r}\|$, and $\hat{r} = \mathbf{r}/r$.

To compute the divergence, we convert to Cartesian coordinates. Let $k = GMm$.

$$\mathbf{F}(x, y, z) = -k\frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{(x^2 + y^2 + z^2)^{3/2}} \tag{4}$$

Note that the denominator is r^3 because $\hat{r}/r^2 = (\mathbf{r}/r)/r^2 = \mathbf{r}/r^3$.

5.1 Calculation of Partial Derivatives

We compute $\frac{\partial F_1}{\partial x}$ where $F_1 = -kx(x^2 + y^2 + z^2)^{-3/2}$.

Using the product rule and the chain rule:

$$\begin{aligned}
\frac{\partial}{\partial x}\left[x(x^2 + y^2 + z^2)^{-3/2}\right] &= (1)(x^2 + y^2 + z^2)^{-3/2} + x\frac{\partial}{\partial x}(x^2 + y^2 + z^2)^{-3/2} \\
&= (x^2 + y^2 + z^2)^{-3/2} + x\left[-\frac{3}{2}(x^2 + y^2 + z^2)^{-5/2} \cdot (2x)\right] \\
&= \frac{1}{(x^2 + y^2 + z^2)^{3/2}} - \frac{3x^2}{(x^2 + y^2 + z^2)^{5/2}}
\end{aligned}$$

Multiplying by the constant $-k$:

$$\frac{\partial F_1}{\partial x} = -k\left[\frac{1}{r^3} - \frac{3x^2}{r^5}\right] \tag{5}$$

By the symmetry of the variables, the derivatives for y and z are:

$$\frac{\partial F_2}{\partial y} = -k\left[\frac{1}{r^3} - \frac{3y^2}{r^5}\right], \quad \frac{\partial F_3}{\partial z} = -k\left[\frac{1}{r^3} - \frac{3z^2}{r^5}\right] \tag{6}$$

5.2 Summation

$$
\begin{aligned}
\nabla \cdot \mathbf{F} &= \sum_{i=1}^{3} \partial_i F_i \\
&= -k \left[\left(\frac{1}{r^3} + \frac{1}{r^3} + \frac{1}{r^3} \right) - \frac{3(x^2 + y^2 + z^2)}{r^5} \right] \\
&= -k \left[\frac{3}{r^3} - \frac{3(r^2)}{r^5} \right] \\
&= -k \left[\frac{3}{r^3} - \frac{3}{r^3} \right] = 0
\end{aligned}
$$

Thus, $\nabla \cdot \mathbf{F} = 0$ for all $\mathbf{r} \neq \mathbf{0}$.

6 Clarification: Direction vs. Divergence

A common confusion arises from conflating the direction of the vector field with its divergence.

Query: The field $\mathbf{F}$ is everywhere directed toward the origin (inward). Why does this not imply $\nabla \cdot \mathbf{F} < 0$?

Mathematical Explanation: The divergence operator $\nabla \cdot \mathbf{F}$ does not measure the direction of the vectors; it measures the change in the density of the vector field flux.

Consider a spherical shell S centered at the origin with inner radius r_1 and outer radius r_2.

1. The field points inward, so flux enters at r_2 and leaves at r_1 (towards the origin).
2. The surface area of a sphere scales as $A(r) = 4\pi r^2$.
3. The magnitude of the field scales as $\|\mathbf{F}\| \propto 1/r^2$.

The total flux Φ through a spherical surface of radius r is proportional to:

$$
\Phi \propto \text{Area} \times \text{Magnitude} \propto r^2 \times \frac{1}{r^2} = \text{Constant}
$$

Because the flux is constant regardless of the radius r, the net flux entering any region bounded by two spheres is exactly equal to the net flux leaving that region. The geometric convergence of the field lines (getting closer together because the area available for them is shrinking) is exactly compensated by the increasing magnitude of the vector field.

Therefore, locally at any point $\mathbf{r} \neq \mathbf{0}$, there is zero net accumulation, and $\nabla \cdot \mathbf{F} = 0$.

~~Lec4~~ Divergence & Curl of a vector field

Till introduction of total derivative, we will assume that "differentiability" of a function of more than one variable at a point P means that all its partial derivatives exist & are continuous at P.

Let $\vec{v}(x,y,z)$ be a differentiable vector function & let v_1, v_2, v_3 be the components of $\vec{v}$. Then, the function

$$\text{div}\,\vec{v} = \nabla\cdot\vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} \quad -(1)$$

is called the divergence of $\vec{v}$.

Symbolic vector : ∇ can be thought of as the symbolic vector $\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$ so that $\nabla\cdot\vec{v} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\cdot\left(v_1\hat{i} + v_2\hat{j} + v_3\hat{k}\right)$
$= (1)$.

Eg 1) Suppose $\vec{v} = x^2z^2\hat{i} - 2y^2z^2\hat{j} + xy^2z\hat{k}$. Find div $\vec{v}$ at $(1,-1,1)$.

$$\nabla\cdot\vec{v} = 2xz^2 + (-4yz^2) + xy^2\Big|_{(1,-1,1)} = 2 + -4\times-1 + 1 = 7.$$

Eg. 2) Given $\phi(x,y,z) = 6x^3y^2z$, a scalar function,

a) find grad ϕ

b) find div(grad ϕ).

Soln :- a) $\left(\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z}\right) = \nabla\phi = (18x^2y^2z, 12x^3yz, 6x^3y^2)$

b) $\nabla\cdot(\nabla\phi) = \text{div}(\text{grad}\,\phi) = \cancel{\frac{\partial}{\partial x}\frac{\partial\phi}{\partial x} + \frac{\partial}{\partial y}\frac{\partial\phi}{\partial y} + \frac{\partial}{\partial y}\frac{\partial\phi}{\partial y}}$

$= (36xy^2z, 12x^3z, 0)$

$\cancel{\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2}}$ Note that we had computed

$$\frac{\partial}{\partial x}\left(\frac{\partial \phi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial \phi}{\partial y}\right) + \frac{\partial}{\partial z}\left(\frac{\partial \phi}{\partial z}\right)$$

$$= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$$

This operator $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is called the Laplacian operator.

You can define it on any twice differentiable scalar function.

Defn:- If a vector function $\vec{v}(P)$ ~~is~~ can be written as gradient of a scalar function f say $\vec{v}(P) = \nabla f(P)$, then the function $f(P)$ is called a potential of $\vec{v}(P)$. Then ~~further,~~ $\vec{v}(P)$ defines a conservative vector field.

~~Eg 3) (Gravitational field)~~

Lec 11 :- Eg 3) (Gravitational field) ~~force of~~ Laplace's eqn

a) The force of attraction between 2 particles at points $P_0 = (x_0, y_0, z_0)$ & $P = (x, y, z)$ given by

$$\vec{p} = -\frac{c}{r^3}\vec{r}$$

where $\vec{r} = \overrightarrow{P_0P}$ can be written as

$$\vec{p} = \nabla f$$

where $f(x,y,z) = \frac{c}{r}$ where $r = \|\vec{r}\|$. That is, $\vec{p} = \text{grad}\left(\frac{c}{r}\right)$.

b) This potential f is a soln of Laplace eqn, ie,

$$\nabla^2 f = 0.$$

Equivalently $\text{div}\,\vec{p} = 0$.

Soln :- $\nabla\left(\frac{c}{r}\right) = \nabla(f) = \left(c\frac{\partial}{\partial x}\left(\frac{1}{r}\right), c\frac{\partial}{\partial y}\left(\frac{1}{r}\right), c\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right)$.

$$r = \|\vec{r}\| = \sqrt{(x-x_0)^2+(y-y_0)^2+(z-z_0)^2}.$$

Let $g(r) = \frac{1}{r}$. Applying Chain Rule, we get

$$\rightarrow \frac{\partial g}{\partial x} = \frac{\partial g}{\partial r}\frac{\partial r}{\partial x} = -\frac{1}{r^2} \times \frac{1 \times 2(x-x_0)}{2\sqrt{(x-x_0)^2+(y-y_0)^2+(z-z_0)^2}}$$

ie, $\frac{\partial r}{\partial x} = \frac{x-x_0}{r}$

$$= -\frac{(x-x_0)}{r^3}.$$

lll^{ly}, we get $\frac{\partial g}{\partial y} = -\left(\frac{y-y_0}{r^3}\right)$ & $\frac{\partial g}{\partial z} = -\frac{(z-z_0)}{r^3}$.

$$\Rightarrow \nabla g = -\left(\frac{x-x_0}{r^3}, \frac{y-y_0}{r^3}, \frac{z-z_0}{r^3}\right) \quad \text{wrt } x, y \,\&\, z.$$

$$\therefore \nabla\left(\frac{c}{r}\right) = -\frac{c}{r^3}(x-x_0, y-y_0, z-z_0)$$

$$= -\frac{c\vec{r}}{r^3} = \vec{p}.$$

Thus, the ~~gr~~ force of attraction $\vec{p} = \nabla\left(\frac{c}{r}\right)$, ~~ie~~ is a the gradient of a scalar function $f(x,y,z) = \frac{c}{r}$. Thus, f is the potential function of $\vec{p}$.

Thus, vector field defined by the force of attraction $\vec{p}$, called the gravitational field is a conservative field.

No energy is lost (or gained) in displacing a body from a pt P to P_0 in the field & back to P.

b) In addition, $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \nabla\cdot(\nabla f) = \nabla\cdot\vec{p}$

To compute this, see that

$$\frac{\partial^2}{\partial x^2}\left(\frac{1}{r}\right) = \frac{\partial}{\partial x}\left(-\frac{(x-x_0)}{r^3}\right) = -(x-x_0)\times -\frac{3}{r^4}\frac{\partial r}{\partial x} + \frac{1}{r^3}\times(-1)$$

$$= \frac{3(x-x_0)}{r^4} \times \frac{x-x_0}{r} - \frac{1}{r^3}$$

$$= \frac{3(x-x_0)^2}{r^5} - \frac{1}{r^3}.$$

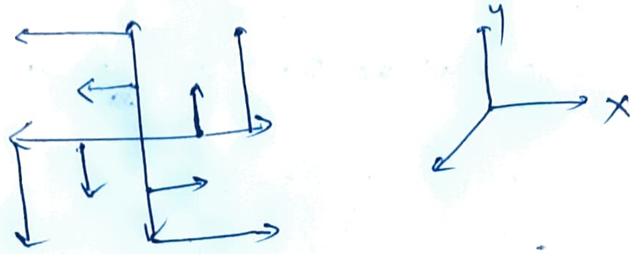
lll^{ly}, $\frac{\partial^2}{\partial y^2}\left(\frac{1}{r}\right) = \frac{3(y-y_0)^2}{r^5} - \frac{1}{r^3}$ & $\frac{\partial^2}{\partial z^2}\left(\frac{1}{r}\right) = \frac{3(z-z_0)^2}{r^5} - \frac{1}{r^3}$

$$\Rightarrow \nabla^2\left(\frac{1}{r}\right) = \frac{3}{r^5}\times r^2 - \left(\frac{1}{r^3}\times 3\right) = 0.$$

ie, $\nabla^2\left(\frac{c}{r}\right) = c\times\nabla^2\left(\frac{1}{r}\right) = 0 = \nabla^2 f$

ie, f satisfies Laplace eqn & $\mathrm{div}(\vec{p}) = \mathrm{div}(\nabla f) = 0$.

Eg3) Find the divergence & curl of the vector field $\vec{f}(x,y) = -y\hat{i} + x\hat{j}$.



$$div(\vec{f}) = \frac{\partial}{\partial x}(-y) + \frac{\partial}{\partial y}(x) = 0$$

$$curl(\vec{f}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y & x & 0 \end{vmatrix} = \hat{i}(0) - \hat{j}(0) + \hat{k}(1-(-1)) = 2\hat{k}$$

Curl($\vec{f}$) given by cross product is along the axis of rotation of the vector field ~~& clockwise direc~~ such that rotation appear clockwise if one looks ~~from~~ in the direction of cross product (by right hand screw rule).

Eg 4) Find curl $\vec{F}$ at $(1,-1,1)$ if $\vec{F} = x^2z^2\hat{i} - 2y^2z^2\hat{j} + xy^2z\hat{k}$.

(Exercise)

~~(Eg5)~~ A vector $\vec{v}$ is called <u>irrotational</u> if $\nabla \times \vec{v} = 0$.

Eg 5) i) Find a, b, c so that $\vec{v} = (-4x-3y+az)\hat{i} + (bx+3y+5z)\hat{j} + (4x+cy+3z)\hat{k}$ is irrotational.

$$curl\,\vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -4x-3y+az & bx+3y+5z & 4x+cy+3z \end{vmatrix} = \hat{i}(c-5) - \hat{j}(4-a) + \hat{k}(b+3) = 0$$

$\Rightarrow a = 4$, $b = -3$, $c = 5$.

~~Now~~ (ii) Show that $\vec{v}$ is a conservative vector field.

Ans:- If $\exists$ scalar field $f: \mathbb{R}^3 \to \mathbb{R}$ st $\nabla f = \vec{v}$, then

$$\frac{\partial f}{\partial x} = -4x - 3y + 4z;$$

$$\frac{\partial f}{\partial y} = -3x + 3y + 5z$$

$$\frac{\partial f}{\partial z} = 4x + 5y + 3z$$

$\Rightarrow f = -2x^2 - 3xy + \cancel{2}4zx + c_1(y,z)$

$= -3xy + \frac{3}{2}y^2 + 5yz + c_2(x,z)$

$= +4xz + 5yz + \frac{3}{2}z^2 + c_3(x,y)$

$\Rightarrow$ By taking common terms from each, we get

$f(x,y,z) = -2x^2 + \frac{3}{2}y^2 + \frac{3}{2}z^2 - 3xy + 5yz + 4xz + c$

~~ie, if curl $\vec{v} = 0$, then $\exists\, f$ s.t. $\nabla f = \vec{v}$~~

~~$\vec{v}$ is irrotational $\Rightarrow$ $\vec{v}$ is conservative~~

~~A continuously differentiable vector function~~

Remark In general, if $\vec{v}$ is an irrotational (differentiable) vector field, then $\exists$ a scalar field f such that $\nabla f = \vec{v}$. (To be checked)

Thm 1) Let $\vec{v}$ be a C^1 ~~vector func~~ continuously differentiable vector function, ~~then~~ that can be written as gradient of a scalar function (f), then its curl is the zero vector.

ie, curl $(\nabla f) = 0$

b) If $\vec{v}$ is twice-~~differentiable~~ continuously differentiable (C^2 smooth) vector function, then

div (curl $\vec{v}$) $= 0$.

Proof :- $\nabla \times \nabla f = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} = \hat{i}\left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y}\right) - \hat{j}\left(\frac{\partial^2 f}{\partial x \partial z} - \frac{\partial^2 f}{\partial z \partial x}\right) + \hat{k}\left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x}\right)$

Note that $\frac{\partial^2 f}{\partial y \partial z} = \frac{\partial^2 f}{\partial z \partial y}$ if all partial derivatives exist & are cts ~~on the~~ at that point.

$\because \frac{\partial^2 f}{\partial y \partial z} = \frac{\partial}{\partial y}(v_3)$ is cont. by assumption

Thus, by c^1 ness of $\vec{v}$, we see that curl (grad f) $= \vec{0}$.

Thus, gravitational field is irrotational too.

Summary : gravitational field is conservative ($\vec{F} = \nabla f$)
irrotational (curl ($\vec{F}$) = 0)
solenoidal (div $\vec{F}$ = 0)

b) $$\nabla \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}$$

$$= \hat{i}\left(\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z} \right) - \hat{j}\left(\frac{\partial v_3}{\partial x} - \frac{\partial v_1}{\partial z} \right) + \hat{k}\left(\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y} \right)$$

$$\nabla \cdot (\nabla \times \vec{v}) = \frac{\partial^2 v_3}{\partial x \partial y} - \frac{\partial^2 v_2}{\partial x \partial z} + \frac{\partial^2 v_1}{\partial y \partial z} - \frac{\partial^2 v_3}{\partial y \partial x} + \frac{\partial^2 v_2}{\partial z \partial x} - \frac{\partial^2 v_1}{\partial z \partial y}$$

Since v_1, v_2 & v_3 have ~~cts~~ 2nd order partial derivatives cts at every pt of the domain,

$$\nabla \cdot (\nabla \times \vec{v}) = 0$$

Lec 10 : Problems

1) Show that $\vec{v} = (-4x - 3y + 4z)\hat{i} + (-3x + 3y + 5z)\hat{j} + (4x + 5y + 3z)\hat{k}$ is a conservative vector field.

Soln: $\frac{\partial f}{\partial x} = -4x - 3y + 4z \Rightarrow f = -2x^2 - 3xy + 4xz + c_1(y, z)$

$= \frac{3y^2}{2} - 3xy + 5yz$

$\Rightarrow f = -2x^2 - 3xy + 4xz + c_1(y, z)$

$f = -3xy \qquad \frac{3y^2}{2} + 5yz + c_2(x, z)$

$= 4xz \qquad 5yz + \frac{3}{2}z^2 + c_3(x,y).$

$\Rightarrow f = -2x^2 + \frac{3}{2}y^2 + \frac{3}{2}z^2 - 3xy + 4xz + 5yz + c$

Remark An easier way is the following result.

Let $\vec{F}$ be a continuously differentiable vector valued function that is irrotational. Then, $\vec{F}$ is conservative.

Proof uses Stokes theorem.

However, the following ~~is true~~. The reverse direction is easier to see. (~~Gradient~~ Conservative fields are irrotational) (2)

~~Suppose $\vec{F}$ is a conservative~~

Prob 2) If a continuously differentiable vector function (on $\mathbb{R}^3$) is the gradient of a scalar function f, then its curl is the zero vector

$$\nabla \times (\nabla f) = \vec{0}$$

Soln :- $\nabla \times (\nabla f) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix}$

$$= \hat{i}\left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \right) - \hat{j}\left(\frac{\partial^2 f}{\partial x \partial z} - \frac{\partial^2 f}{\partial z \partial x} \right) + \hat{k}\left(\frac{\partial^2 f}{\partial x \partial z} - \frac{\partial^2 f}{\partial z \partial x} \right)$$

Remark :-

Since all second order partial derivatives of f are everywhere continuous, their order can be interchanged $\Rightarrow \nabla \times (\nabla f) = \vec{0}$.

Refer Theorem 8.12 Apostol (Calculus Vol II)

Summary : $\nabla \times \vec{F} = \vec{0} \iff \vec{F}$ is conservative (provided $\vec{F}$ is $C^1(\mathbb{R}^3)$)

Prob 3) Let $\vec{V}$ be a vector field whose first & second order partial derivatives are continuous on $\mathbb{R}^3$. Then, $\operatorname{div}(\operatorname{curl} \vec{V}) = 0$

Soln :- Done in class or refer Kreyszig

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Exercise: Assuming sufficient differentiability & continuity of partial derivatives of required order, show that

1) $\operatorname{div}(\vec{u} \times \vec{v}) = \vec{v} \cdot \operatorname{curl} \vec{u} - \vec{u} \cdot (\operatorname{curl} \vec{v})$.

2) a) $\nabla(fg) = f\nabla g + g\nabla f$

b) $\nabla\left(\frac{f}{g}\right) = \frac{g\nabla f - f\nabla g}{g^2}$

3) $\operatorname{div}(f\vec{v}) = f \operatorname{div} \vec{v} + \vec{v} \cdot \nabla f$

4) Show that if $\phi : \mathbb{R}^3 \to \mathbb{R}$ is a C^2 scalar field ~~so~~ such that $\nabla^2 \phi = 0$, then $\nabla\phi$ is solenoidal & irrotational.

Soln :- $\nabla^2 \phi = \nabla \cdot (\nabla\phi) = \operatorname{div}(\nabla\phi) = 0 \Rightarrow \nabla\phi$ is solenoidal

$\nabla\phi$ is clearly conservative $\Rightarrow \nabla\phi$ is irrotational

Eg. Gravitational field